American International University-Bangladesh (AIUB)

Faculty of Science & Technology

Department of Computer Science

Object Oriented Programming – 2

Course Code: CSC2210

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Section: K

PROJECT REPORT ON

Event Management System

A Hall Booking and Event Service Management Application for Bangladesh

Supervised By

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Letter to the Professor

Dear Professor,

Dr. Md. Iftekharul Mobin

We would like to express our sincere gratitude for your kind guidance and continuous support during our Object Oriented Programming – 2 course. The lessons you have taught us, both in theory classes and in lab sessions, have helped us understand C# programming, object-oriented design, and database integration in a clearer way.

Our team has chosen the project titled **"Event Management System"**, which is a Windows Forms desktop application that helps customers in Bangladesh book event halls such as Sena Kunja, Army Golf Club, Sena Malancha, Dhaka Club, and several community and convention centres. The system also allows event service providers to offer extra services like decoration, catering, photography, stage setup and event planning. We have tried our best to apply the OOP concepts, GUI design ideas, and database knowledge that we learned from your course in this project.

We humbly request you to evaluate our work and provide your valuable feedback so that we can improve further. We have given our best effort, and we hope that this report and our project meet your expectations.

Thank you very much for your time and guidance.

Best regards,

Group Members

Tasnim Mahmud Jeba
Mehrah Khan

Halimatus Sadia

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1. Introduction :

Booking an event hall in Bangladesh is still mostly a manual process. Customers usually have to call different halls, visit them physically, compare prices, check capacity, and ask about date availability before making a final decision. This process takes time and can be confusing for both customers and hall owners. The Event Management System is a C# Windows Forms desktop application designed to make this process easier and more organized. The system brings hall booking, event information, service provider records, payments, and reviews into one platform. Customers can search for halls by location, price range, capacity, event type, and available date. They can also book halls for events such as weddings, birthdays, conferences, and receptions. The system supports three types of login users: Customers, Admins / Hall Owners, and a Super Admin. Customers can browse halls and make bookings. Admins or hall owners can manage hall details, prices, discounts, images, and busy dates. The Super Admin can manage the overall system, including users, halls, bookings, payments, and reviews. Service providers such as decorators, caterers, photographers, and event planners are stored as database records and managed by the Admin or Super Admin. They are not separate login accounts. This application is built using C# Windows Forms for the user interface and Microsoft SQL Server for the database. It uses ADO.NET to connect the application with the database and store important information such as customer data, hall details, bookings, payments, and reviews. The project also applies object-oriented programming concepts such as classes, objects, encapsulation, inheritance, and exception handling to keep the system structured, reusable, and reliable.

2. Case Study :

Mr. Rahman is planning his daughter's wedding in Mirpur for around 500 guests. He opens the Event Management System and signs up as a customer (Role = Customer). After login he reaches the Customer Dashboard and uses the Hall Search Page to filter by location, capacity and price. The system shows matching halls including Sena Kunja and Sena Malancha. He clicks Sena Kunja, sees images, prices, availability and busy dates, and presses Book Now. On the Booking Page he picks an event date, selects "Wedding" from EventTypes, fills in start time, end time and guest count, and confirms. The system inserts a new row in Bookings with the status Pending and shows the Payment Page, where Mr. Rahman pays the total amount. The booking now moves to the hall owner — who logs in with the Admin / Hall Owner role — and approves it from the Hall Owner Dashboard. To arrange the decoration and catering, the Super Admin opens the EventAssignments tab on the Super Admin Dashboard, picks the booking, and chooses a service provider company (for example, "Royal Decor & Catering") from the ServiceProviders table. A short note can be added in the Notes field. The service provider company itself does not log in — its details are stored as data and managed by the admin. After the wedding, Mr. Rahman returns to give a 5-star review.

3. Functional Requirements

Based on the case study above, the main functional requirements of the Event Management System are listed below. These requirements were used as the base for designing the database, the user interface, and the C# classes of the project.

1. User registration and login.
2. Three login roles only — Customer, Admin / Hall Owner and Super Admin — stored in the Roles table.
3. Hall listing with name, location, price, capacity, availability, image and owner.
4. Search and filtering for customers (location, event type, price, capacity, date).
5. Booking calendar showing available and busy dates / times for each hall.
6. Bookings for weddings, birthdays, conferences, receptions and other event types.
7. Payments per booking with multiple payment methods.
8. Customer reviews with 1–5 star ratings.
9. Hall owner / admin features: add, update, delete halls and approve bookings.
10. Service provider companies are stored in the ServiceProviders table (data only — no login). The Admin / Super Admin assigns them to a booking through the EventAssignments table.
11. Super Admin can manage every table in the database.
12. Built with C# Windows Forms (UI) and Microsoft SQL Server using ADO.NET.

4. ER Diagram

The Entity Relationship (ER) Diagram of our Event Management System is shown below. It clearly identifies the main entities of the system and the relationships between them.

Main Entities

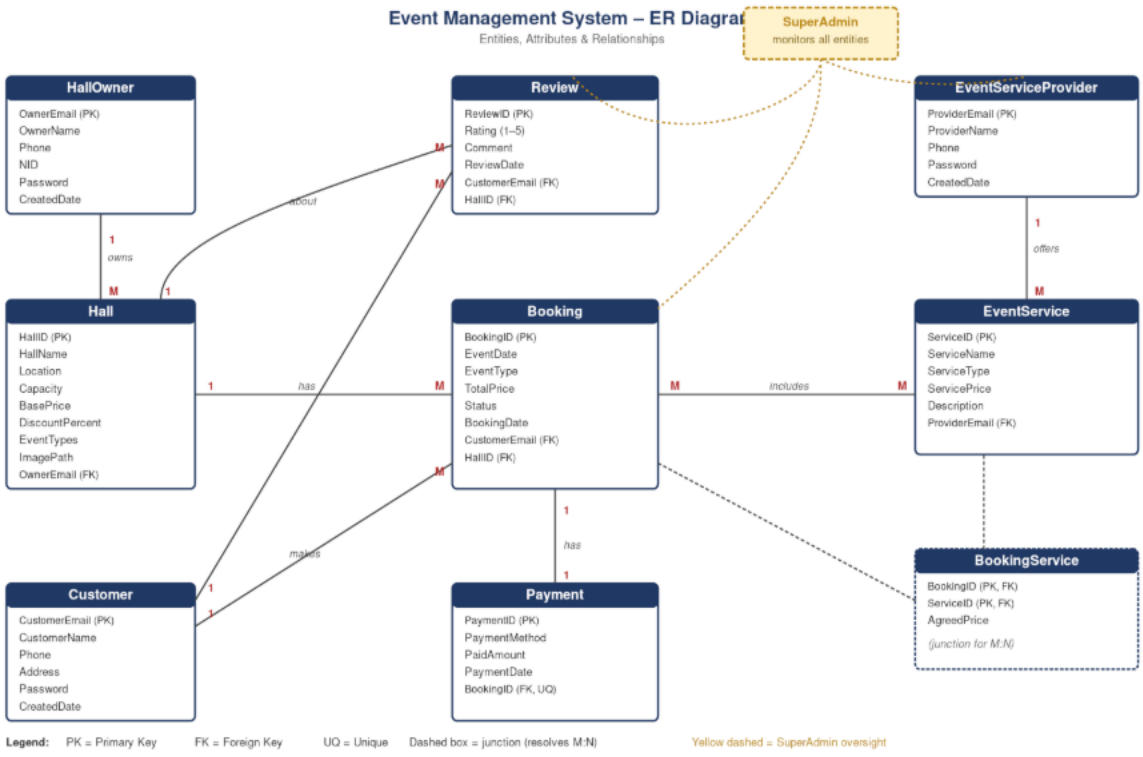
- Customer — books halls and gives reviews.
- HallOwner — owns one or more halls and manages bookings.
- Hall — the physical event hall that customers book.
- Booking — the record created when a customer books a hall.
- Payment — payment information for a booking.
- Review — rating and feedback given by a customer.
- EventServiceProvider — companies that offer extra event services.
- EventService — the actual service (decoration, catering, photography, stage setup, event planning) offered by a provider.
- SuperAdmin — controls the whole system.

Main Relationships

- A HallOwner owns many Halls (1 : M).
- A Customer makes many Bookings (1 : M).
- A Hall receives many Bookings (1 : M).
- A Booking has one Payment (1 : 1).

- A Customer writes many Reviews (1 : M).
- A Hall receives many Reviews (1 : M).
- An EventServiceProvider offers many EventServices (1 : M).
- A Booking can include many EventServices (M : N — handled through a junction structure inside the Booking and EventService relation).
- A SuperAdmin monitors everything (logical, system-wide relationship).

ER Diagram:



5. Normalization and Finalization

In this section we go through the normalization process for each major relationship in the Event Management System. The goal is to remove repeating groups, partial dependencies, and transitive dependencies, so that the final database is clean and easy to maintain.

Owns: HallOwner — Hall (1 : M)

UNF: ownerName, ownerEmail, ownerPhone, ownerNID, ownerPassword, hallID, hallName, location, capacity, basePrice, eventTypes, description.

1NF: ownerName, ownerEmail, ownerPhone, ownerNID, ownerPassword, hallID, hallName, location, capacity, basePrice, eventTypes, description.

2NF:

1. 1. ownerEmail (PK), ownerName, ownerPhone, ownerNID, ownerPassword.
2. 2. hallID (PK), hallName, location, capacity, basePrice, eventTypes, description, ownerEmail (FK).

3NF: Same as 2NF.

Books: Customer — Booking (1 : M)

UNF: customerEmail, customerName, customerPhone, customerAddress, customerPassword, bookingID, eventDate, eventType, totalPrice, status.

1NF: customerEmail, customerName, customerPhone, customerAddress, customerPassword, bookingID, eventDate, eventType, totalPrice, status.

2NF:

3. 1. customerEmail (PK), customerName, customerPhone, customerAddress, customerPassword.
4. 2. bookingID (PK), eventDate, eventType, totalPrice, status, customerEmail (FK).

3NF: Same as 2NF.

Receives: Hall — Booking (1 : M)

UNF: hallID, hallName, location, capacity, basePrice, bookingID, eventDate, eventType, totalPrice, status.

1NF: hallID, hallName, location, capacity, basePrice, bookingID, eventDate, eventType, totalPrice, status.

2NF:

5. 1. hallID (PK), hallName, location, capacity, basePrice.
6. 2. bookingID (PK), eventDate, eventType, totalPrice, status, hallID (FK).

3NF: Same as 2NF.

Pays: Booking — Payment (1 : 1)

UNF: bookingID, eventDate, totalPrice, status, paymentID, paymentMethod, paidAmount, paymentDate.

1NF: bookingID, eventDate, totalPrice, status, paymentID, paymentMethod, paidAmount, paymentDate.

2NF:

7. 1. bookingID (PK), eventDate, totalPrice, status.
8. 2. paymentID (PK), paymentMethod, paidAmount, paymentDate, bookingID (FK).

3NF: Same as 2NF.

Writes: Customer — Review (1 : M)

UNF: customerEmail, customerName, customerPhone, reviewID, rating, comment, reviewDate, hallID.

1NF: customerEmail, customerName, customerPhone, reviewID, rating, comment, reviewDate, hallID.

2NF:

9. 1. customerEmail (PK), customerName, customerPhone.
10. 2. reviewID (PK), rating, comment, reviewDate, hallID (FK), customerEmail (FK).

3NF: Same as 2NF.

Provides: EventServiceProvider — EventService (1 : M)

UNF: providerEmail, providerName, providerPhone, providerPassword, serviceID, serviceName, serviceType, servicePrice, description.

1NF: providerEmail, providerName, providerPhone, providerPassword, serviceID, serviceName, serviceType, servicePrice, description.

2NF:

11. 1. providerEmail (PK), providerName, providerPhone, providerPassword.
12. 2. serviceID (PK), serviceName, serviceType, servicePrice, description, providerEmail (FK).

3NF: Same as 2NF.

Includes: Booking — EventService (M : N)

UNF: bookingID, totalPrice, serviceID, serviceName, servicePrice.

1NF: bookingID, totalPrice, serviceID, serviceName, servicePrice.

2NF:

13. 1. bookingID (PK), totalPrice.
14. 2. serviceID (PK), serviceName, servicePrice.
15. 3. BookingService: bookingID (FK), serviceID (FK), agreedPrice — composite key (bookingID, serviceID).

3NF: Same as 2NF.

Controls: SuperAdmin — System

UNF: adminID, adminName, adminEmail, adminPassword, lastLogin.

1NF: adminID, adminName, adminEmail, adminPassword, lastLogin.

2NF:

16. 1. adminID (PK), adminName, adminEmail, adminPassword, lastLogin.

3NF: Same as 2NF.

Finalization

After applying UNF → 1NF → 2NF → 3NF on each relationship, we combine all the resulting tables into a single clean database design. This gives us the final list of tables shown in the next subsection.

Final Tables

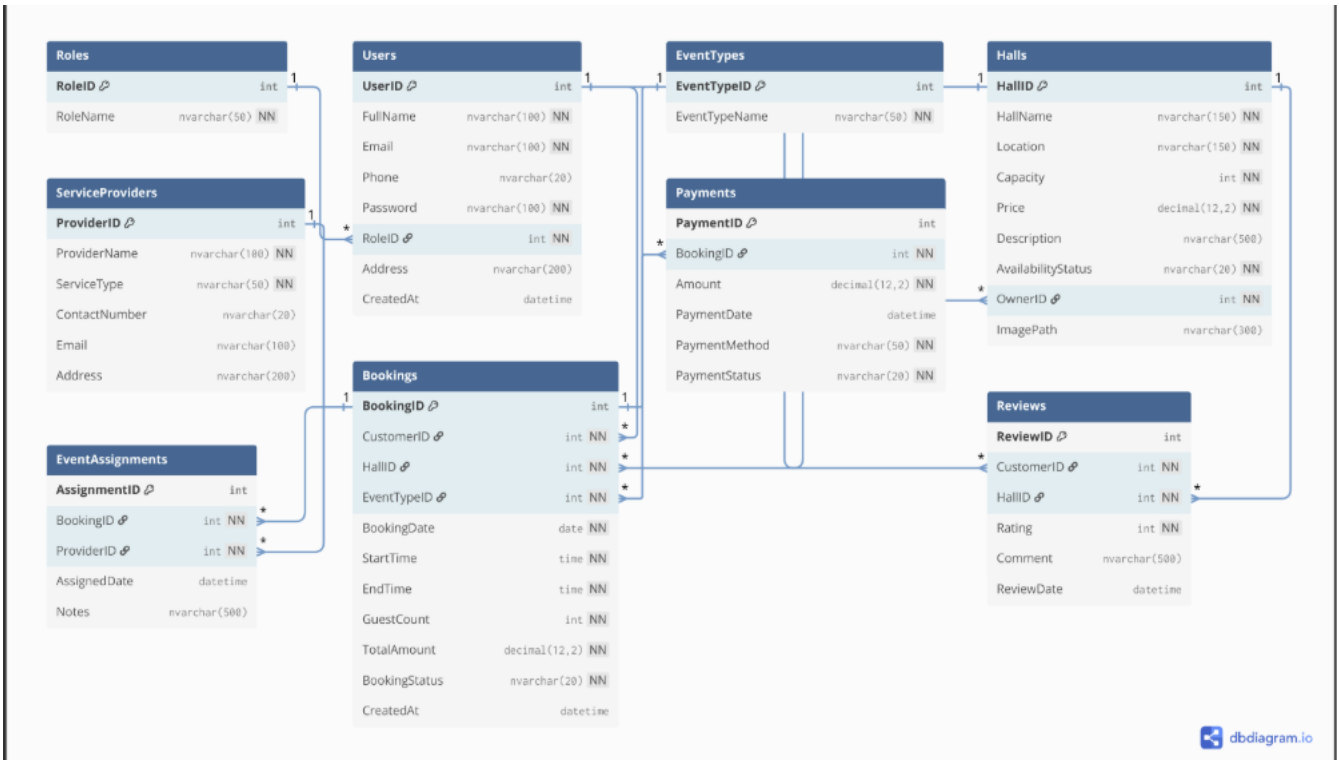
17. Customer (CustomerEmail PK, CustomerName, Phone, Address, Password).
18. HallOwner (OwnerEmail PK, OwnerName, Phone, NID, Password).
19. Hall (HallID PK, HallName, Location, Capacity, BasePrice, EventTypes, Description, ImagePath, OwnerEmail FK).

20. Booking (BookingID PK, EventDate, EventType, TotalPrice, Status, CustomerEmail FK, HallID FK).
21. Payment (PaymentID PK, PaymentMethod, PaidAmount, PaymentDate, BookingID FK).
22. Review (ReviewID PK, Rating, Comment, ReviewDate, CustomerEmail FK, HallID FK).
23. EventServiceProvider (ProviderEmail PK, ProviderName, Phone, Password).
24. EventService (ServiceID PK, ServiceName, ServiceType, ServicePrice, Description, ProviderEmail FK).
25. BookingService (BookingID FK, ServiceID FK, AgreedPrice — Composite PK).
26. SuperAdmin (AdminID PK, AdminName, AdminEmail, AdminPassword, LastLogin).

6. Database Schema

The database schema diagram below shows all the tables, their columns, primary keys, and foreign key relationships. This schema was drawn based on the final tables we got after normalization.

Schema Diagram :



Short Explanation of the Schema

The Customer, HallOwner, EventServiceProvider, and SuperAdmin tables store the four types of users in the system. The Hall table is connected to the HallOwner table through OwnerEmail. The Booking table connects each customer to a specific hall and stores the event date, event type, and status (pending / approved / rejected / completed).

Each booking has exactly one Payment record connected through BookingID. The Review table stores customer feedback for each hall. The EventService table stores the services offered by providers, and the BookingService junction table connects bookings with the services chosen by the customer. This many-to-many relationship is the reason we needed an extra junction table after 3NF.

7. SQL Database Queries

All the SQL queries below were written for Microsoft SQL Server. They were tested in SQL Server Management Studio (SSMS) before being added to this report. Tables were created in this order so that foreign keys never reference a table that does not yet exist.

Create Database

```
CREATE DATABASE EventManagementDB;
GO

USE EventManagementDB;
GO
```

1. Customer Table

```
CREATE TABLE Customer (
    CustomerEmail   NVARCHAR(100) PRIMARY KEY,
    CustomerName    NVARCHAR(100) NOT NULL,
    Phone           NVARCHAR(20)  NOT NULL,
    Address         NVARCHAR(200),
    Password        NVARCHAR(100) NOT NULL,
    CreatedDate     DATETIME      NOT NULL DEFAULT GETDATE()
);
```

2. HallOwner Table

```
CREATE TABLE HallOwner (
    OwnerEmail      NVARCHAR(100) PRIMARY KEY,
    OwnerName       NVARCHAR(100) NOT NULL,
    Phone           NVARCHAR(20)  NOT NULL,
    NID             NVARCHAR(30),
    Password        NVARCHAR(100) NOT NULL,
    CreatedDate     DATETIME      NOT NULL DEFAULT GETDATE()
);
```

3. Hall Table

```
CREATE TABLE Hall (
    HallID          INT IDENTITY(1,1) PRIMARY KEY,
    HallName        NVARCHAR(150) NOT NULL,
    Location        NVARCHAR(150) NOT NULL,
```

```
    Capacity        INT           NOT NULL CHECK (Capacity > 0),
    BasePrice       DECIMAL(12,2) NOT NULL CHECK (BasePrice >= 0),
    DiscountPercent DECIMAL(5,2)  NOT NULL DEFAULT 0
                       CHECK (DiscountPercent BETWEEN 0 AND 100),
    EventTypes      NVARCHAR(200),
    Description     NVARCHAR(500),
    ImagePath       NVARCHAR(300),
    OwnerEmail      NVARCHAR(100) NOT NULL,
    CONSTRAINT FK_Hall_Owner FOREIGN KEY (OwnerEmail)
        REFERENCES HallOwner(OwnerEmail)
);
```

4. EventServiceProvider Table

```
CREATE TABLE EventServiceProvider (
    ProviderEmail   NVARCHAR(100) PRIMARY KEY,
    ProviderName    NVARCHAR(100) NOT NULL,
    Phone           NVARCHAR(20)  NOT NULL,
    Password        NVARCHAR(100) NOT NULL,
    CreatedDate     DATETIME      NOT NULL DEFAULT GETDATE()
);
```

5. EventService Table

```
CREATE TABLE EventService (
    ServiceID       INT IDENTITY(1,1) PRIMARY KEY,
    ServiceName     NVARCHAR(100) NOT NULL,
    ServiceType     NVARCHAR(50)  NOT NULL,
    ServicePrice    DECIMAL(12,2) NOT NULL CHECK (ServicePrice >= 0),
    Description     NVARCHAR(500),
    ProviderEmail   NVARCHAR(100) NOT NULL,
    CONSTRAINT FK_Service_Provider FOREIGN KEY (ProviderEmail)
        REFERENCES EventServiceProvider(ProviderEmail)
);
```

6. Booking Table

```
CREATE TABLE Booking (
    BookingID       INT IDENTITY(1,1) PRIMARY KEY,
    EventDate       DATE          NOT NULL,
    EventType       NVARCHAR(50)  NOT NULL,
    TotalPrice      DECIMAL(12,2) NOT NULL CHECK (TotalPrice >= 0),
    Status          NVARCHAR(20)  NOT NULL DEFAULT 'Pending'
```

```
                        CHECK (Status IN
('Pending','Approved','Rejected','Completed','Cancelled')),
    BookingDate     DATETIME      NOT NULL DEFAULT GETDATE(),
    CustomerEmail   NVARCHAR(100) NOT NULL,
    HallID          INT           NOT NULL,
    CONSTRAINT FK_Booking_Customer FOREIGN KEY (CustomerEmail)
        REFERENCES Customer(CustomerEmail),
    CONSTRAINT FK_Booking_Hall     FOREIGN KEY (HallID)
        REFERENCES Hall(HallID)
);
```

7. BookingService (junction) Table

```
CREATE TABLE BookingService (
    BookingID       INT NOT NULL,
    ServiceID       INT NOT NULL,
    AgreedPrice     DECIMAL(12,2) NOT NULL CHECK (AgreedPrice >= 0),
    CONSTRAINT PK_BookingService PRIMARY KEY (BookingID, ServiceID),
    CONSTRAINT FK_BS_Booking FOREIGN KEY (BookingID)
        REFERENCES Booking(BookingID),
    CONSTRAINT FK_BS_Service FOREIGN KEY (ServiceID)
        REFERENCES EventService(ServiceID)
);
```

8. Payment Table

```
CREATE TABLE Payment (
    PaymentID       INT IDENTITY(1,1) PRIMARY KEY,
    PaymentMethod   NVARCHAR(50)  NOT NULL
                        CHECK (PaymentMethod IN
('Cash','Card','bKash','Nagad','Rocket','Bank')),
    PaidAmount      DECIMAL(12,2) NOT NULL CHECK (PaidAmount >= 0),
    PaymentDate     DATETIME      NOT NULL DEFAULT GETDATE(),
    BookingID       INT           NOT NULL UNIQUE,
    CONSTRAINT FK_Payment_Booking FOREIGN KEY (BookingID)
        REFERENCES Booking(BookingID)
);
```

9. Review Table

```
CREATE TABLE Review (
    ReviewID        INT IDENTITY(1,1) PRIMARY KEY,
    Rating          INT           NOT NULL CHECK (Rating BETWEEN 1 AND 5),
    Comment         NVARCHAR(500),
    ReviewDate      DATETIME      NOT NULL DEFAULT GETDATE(),
    CustomerEmail   NVARCHAR(100) NOT NULL,
```

```
    HallID          INT           NOT NULL,
    CONSTRAINT FK_Review_Customer FOREIGN KEY (CustomerEmail)
        REFERENCES Customer(CustomerEmail),
    CONSTRAINT FK_Review_Hall     FOREIGN KEY (HallID)
        REFERENCES Hall(HallID)
);
```

10. SuperAdmin Table

```
CREATE TABLE SuperAdmin (
    AdminID         INT IDENTITY(1,1) PRIMARY KEY,
    AdminName       NVARCHAR(100) NOT NULL,
    AdminEmail      NVARCHAR(100) NOT NULL UNIQUE,
    AdminPassword   NVARCHAR(100) NOT NULL,
    LastLogin       DATETIME      NULL
);
```

Sample SELECT Queries Used in the Project

```
-- Get all halls in a given location
SELECT * FROM Hall WHERE Location LIKE '%Mirpur%';

-- Get all bookings of a customer
SELECT b.BookingID, h.HallName, b.EventDate, b.EventType, b.TotalPrice,
b.Status
FROM Booking b
INNER JOIN Hall h ON b.HallID = h.HallID
WHERE b.CustomerEmail = 'rahman@example.com';

-- Average rating for each hall
SELECT h.HallName, AVG(CAST(r.Rating AS DECIMAL(3,2))) AS AvgRating
FROM Review r INNER JOIN Hall h ON r.HallID = h.HallID
GROUP BY h.HallName;

-- Approve a pending booking
UPDATE Booking SET Status = 'Approved' WHERE BookingID = 101 AND Status =
'Pending';
```

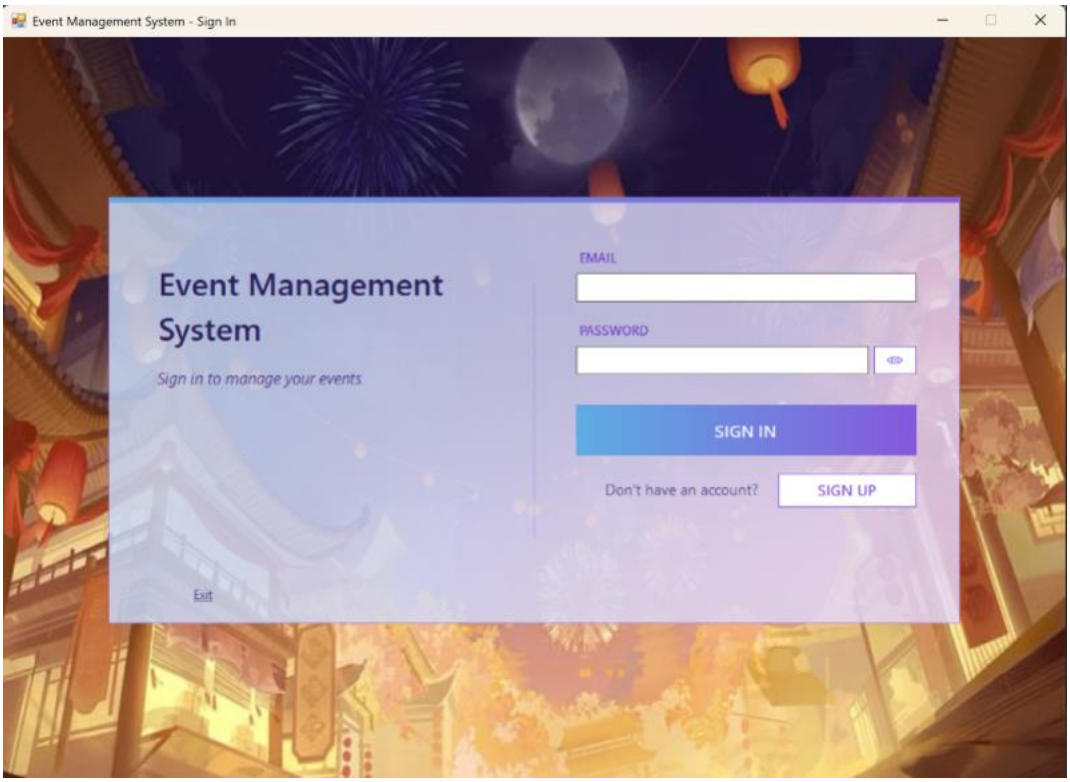
8. User Interface Transition

In this section we describe the major forms (windows) of the Event Management System and how the user moves from one form to another. Screenshots of the actual application have been left as placeholders.

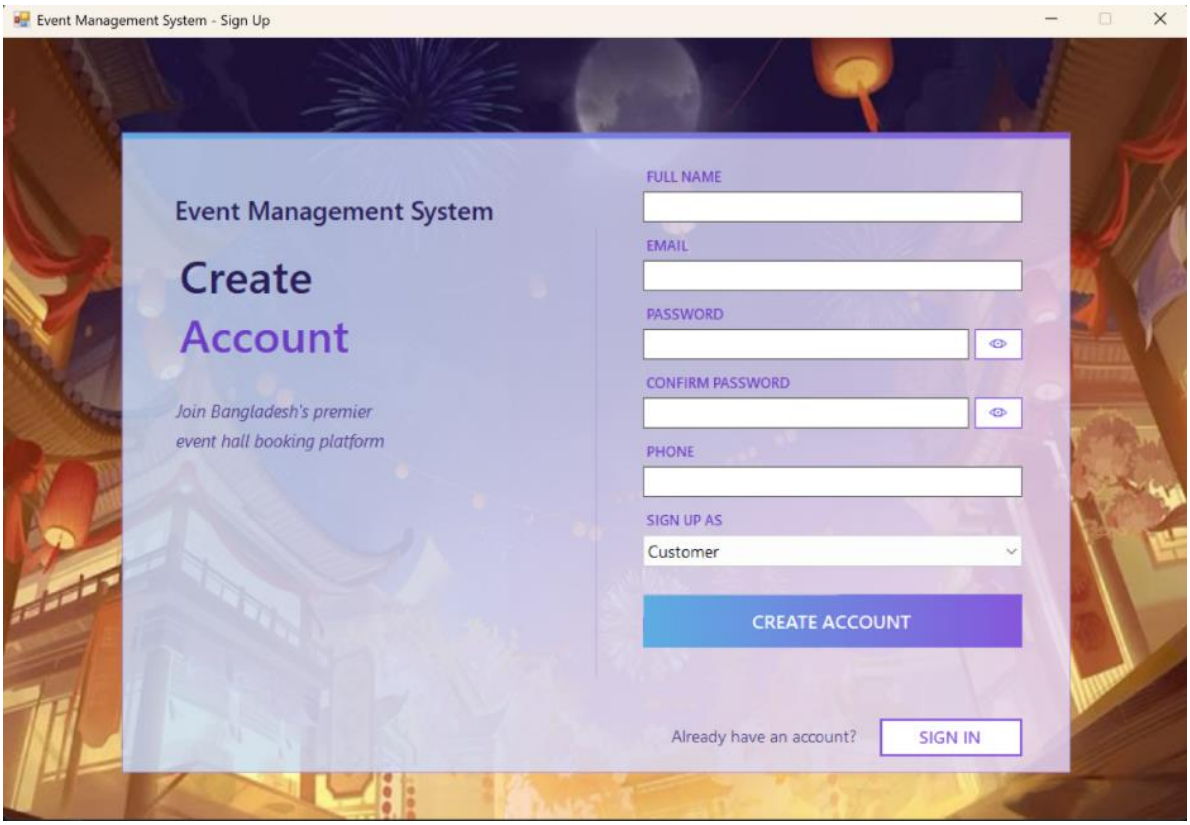
When the app starts, the Login Page opens. After login the user is sent to the dashboard for their role: Customer, Hall Owner / Admin or Super Admin.

8.1 Login Page

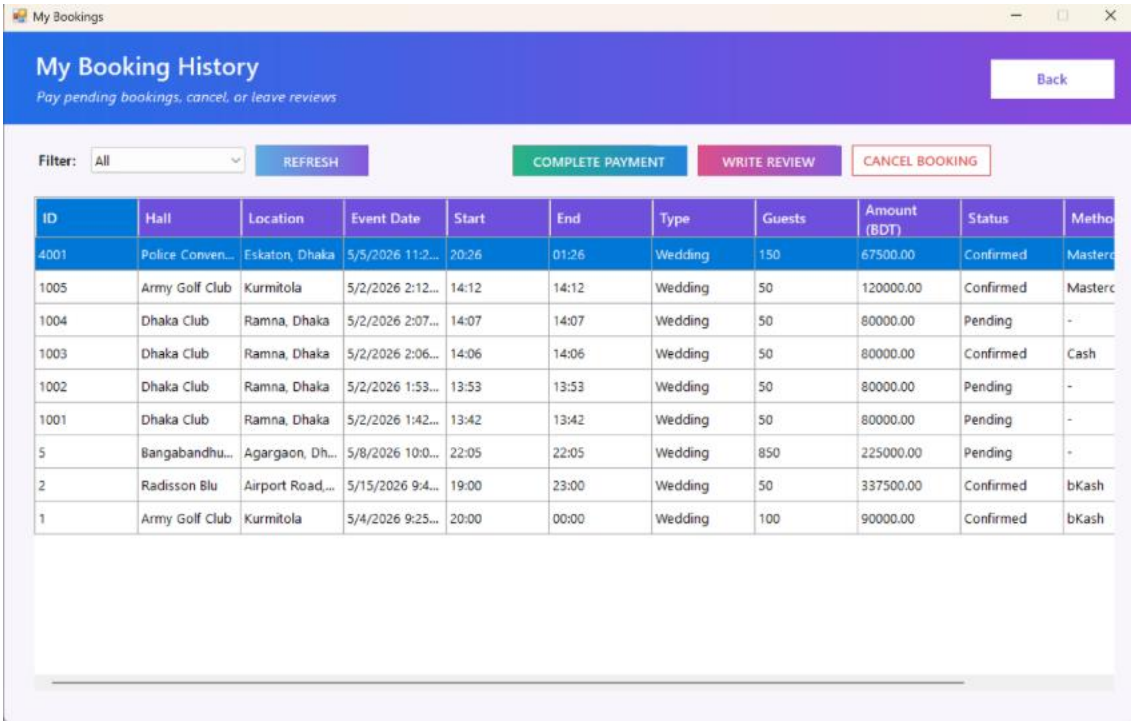
When the app starts, the Login Page opens. After login the user is sent to the dashboard for their role: Customer, Hall Owner / Admin or Super Admin.



8.2 Signup Page

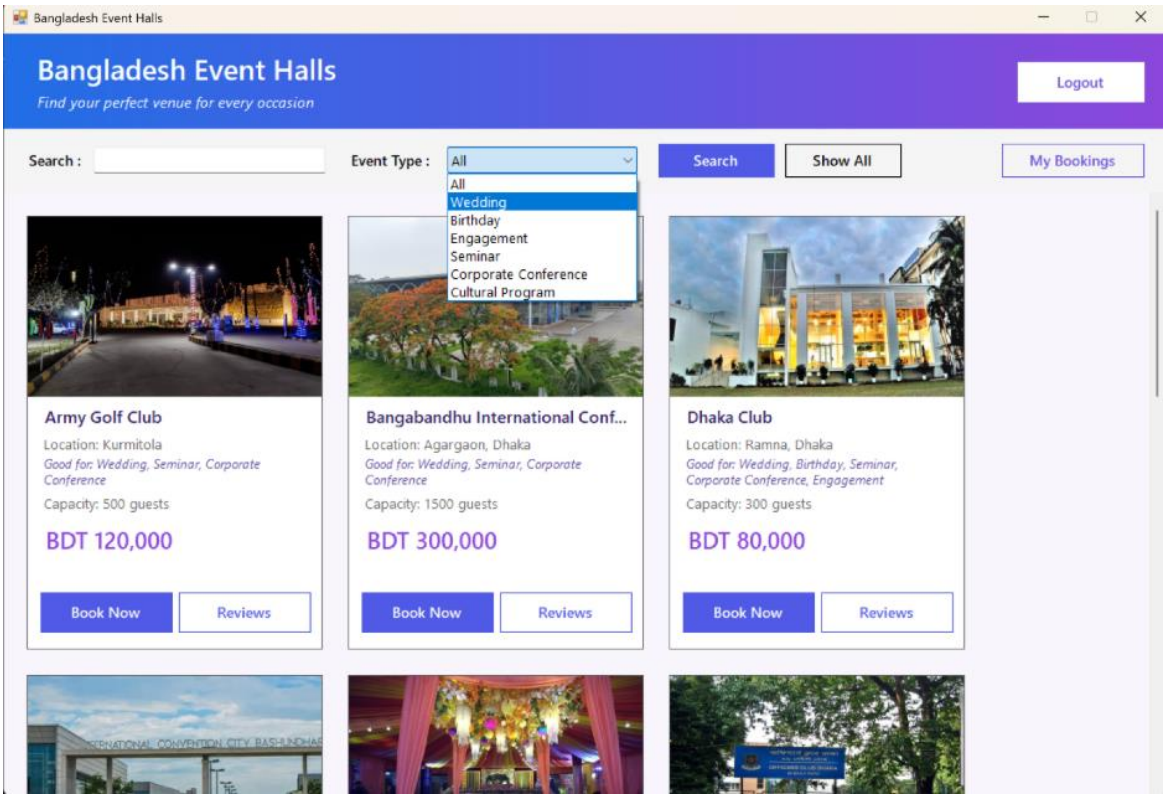


8.3 Customer Dashboard

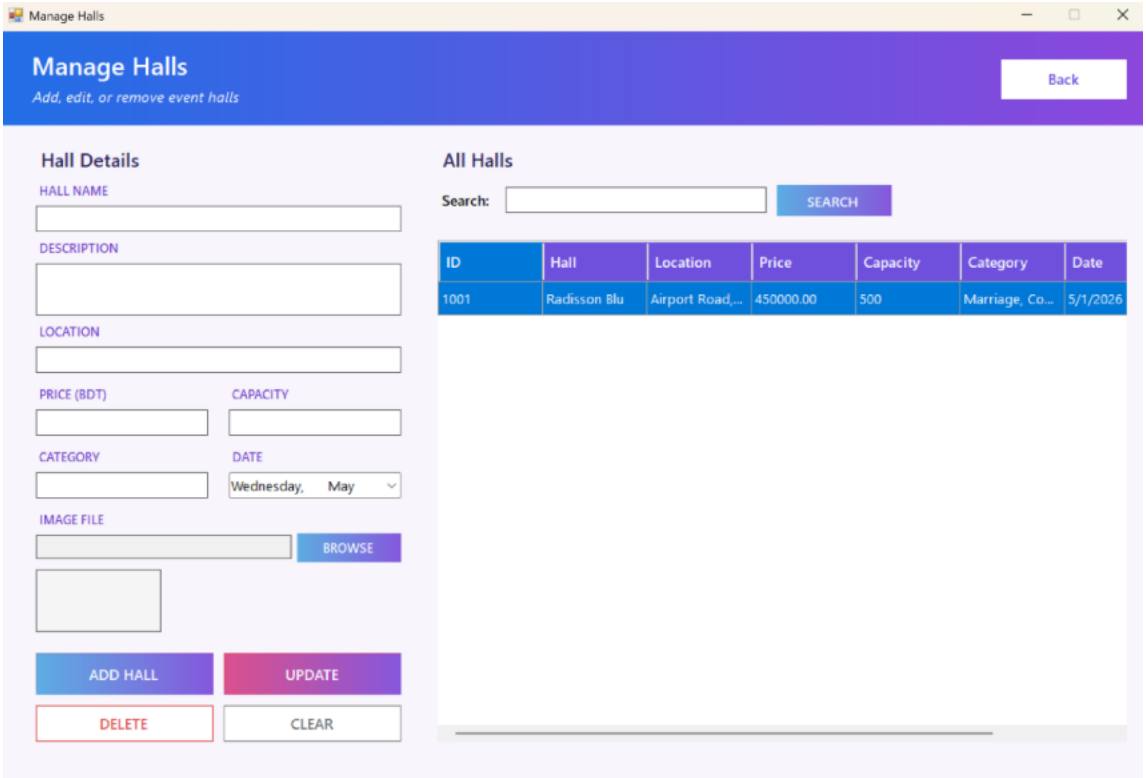


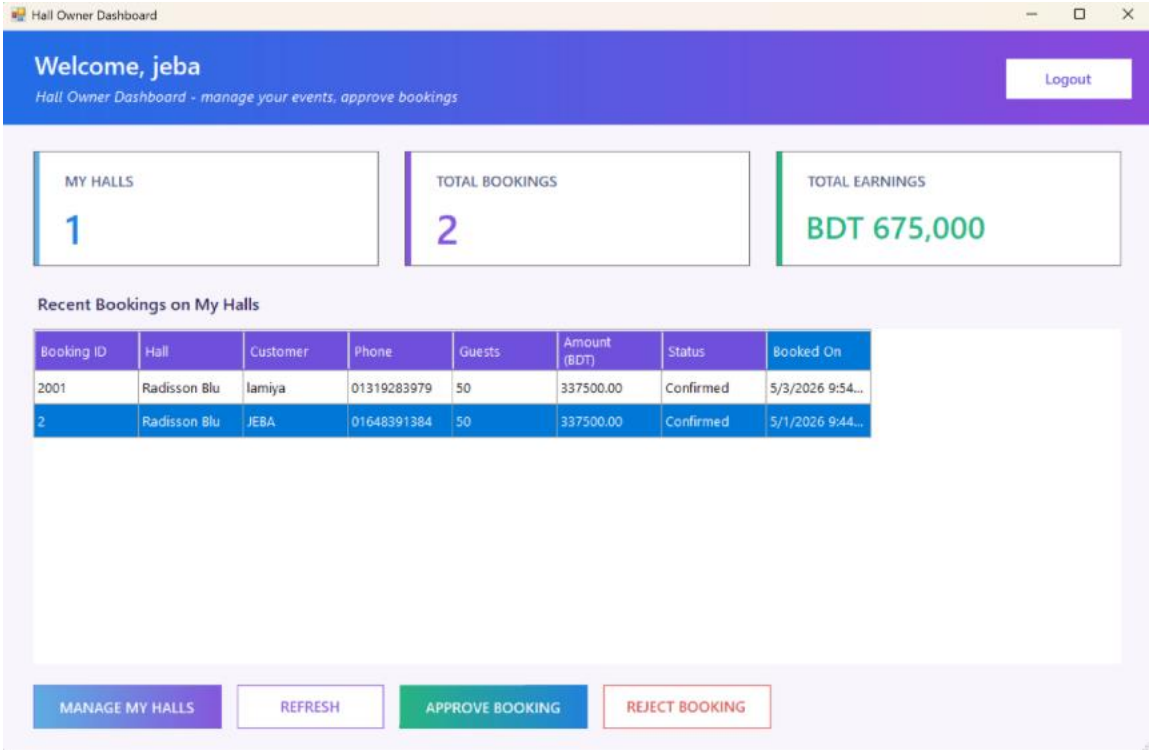
ID	Hall	Location	Event Date	Start	End	Type	Guests	Amount (BDT)	Status	Metho
4001	Police Conven...	Eskaton, Dhaka	5/5/2026 11:2...	20:26	01:26	Wedding	150	67500.00	Confirmed	Masterc
1005	Army Golf Club	Kurmitola	5/2/2026 2:12...	14:12	14:12	Wedding	50	120000.00	Confirmed	Masterc
1004	Dhaka Club	Ramna, Dhaka	5/2/2026 2:07...	14:07	14:07	Wedding	50	80000.00	Pending	-
1003	Dhaka Club	Ramna, Dhaka	5/2/2026 2:06...	14:06	14:06	Wedding	50	80000.00	Confirmed	Cash
1002	Dhaka Club	Ramna, Dhaka	5/2/2026 1:53...	13:53	13:53	Wedding	50	80000.00	Pending	-
1001	Dhaka Club	Ramna, Dhaka	5/2/2026 1:42...	13:42	13:42	Wedding	50	80000.00	Pending	-
5	Bangabandhu...	Agargaon, Dh...	5/8/2026 10:0...	22:05	22:05	Wedding	850	225000.00	Pending	-
2	Radisson Blu	Airport Road,...	5/15/2026 9:4...	19:00	23:00	Wedding	50	337500.00	Confirmed	bKash
1	Army Golf Club	Kurmitola	5/4/2026 9:25...	20:00	00:00	Wedding	100	90000.00	Confirmed	bKash

8.4 Hall Search Page

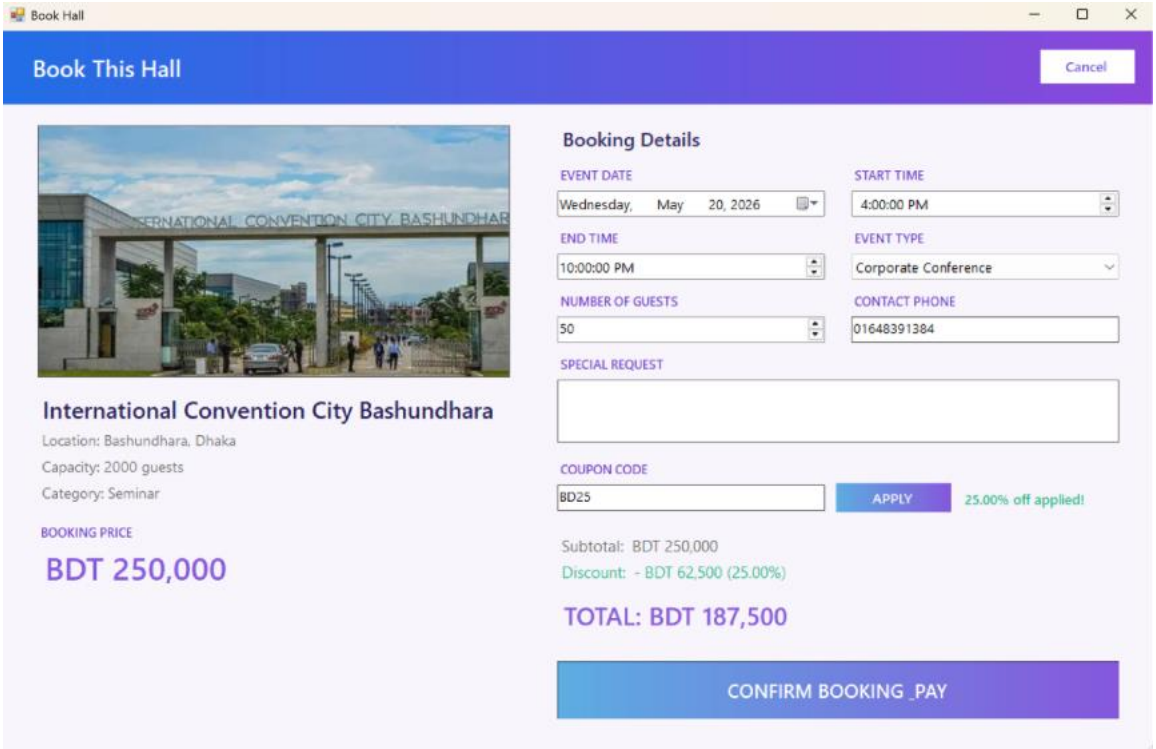


8.5 Admin/ Hall Owner Dashboard

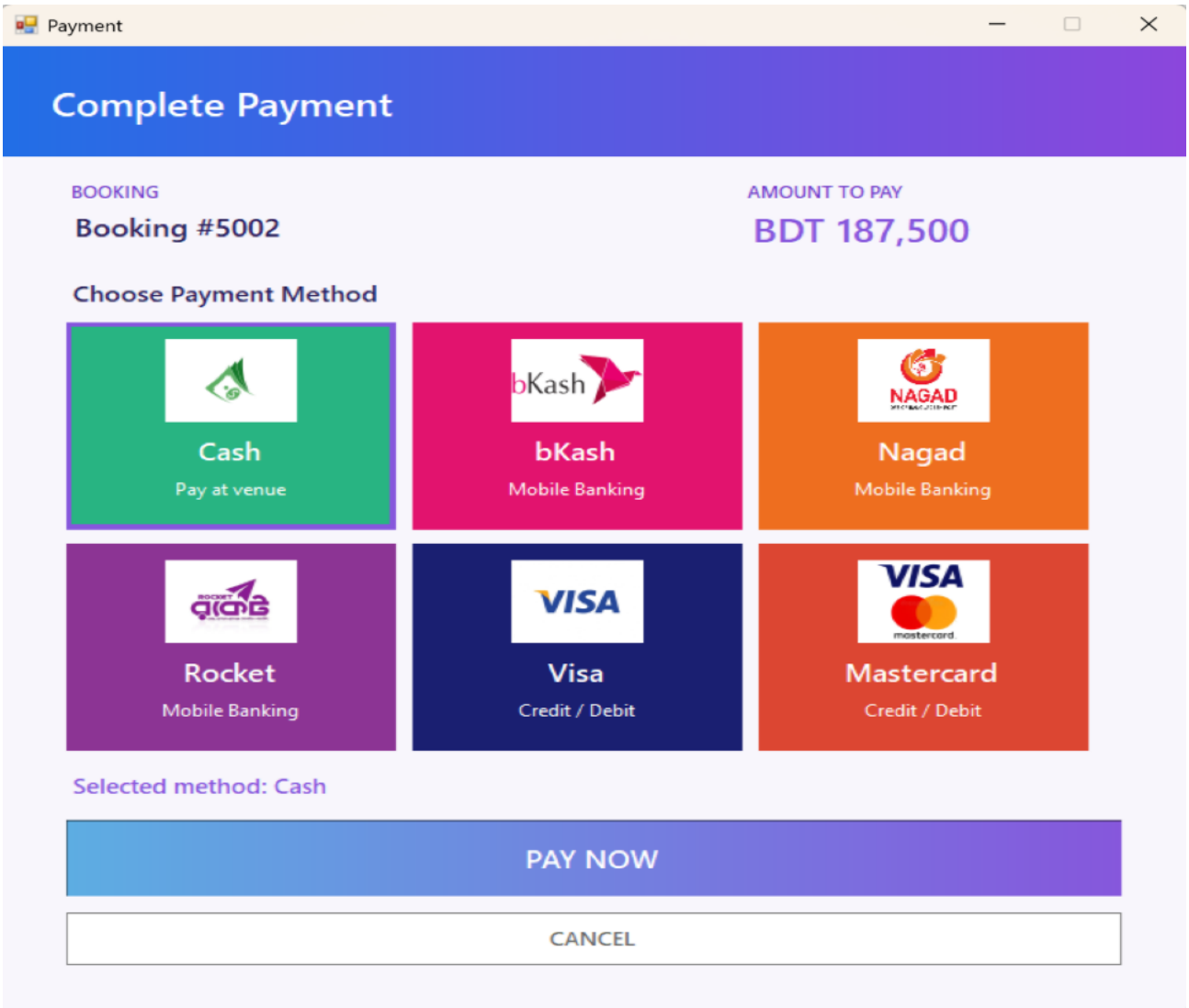




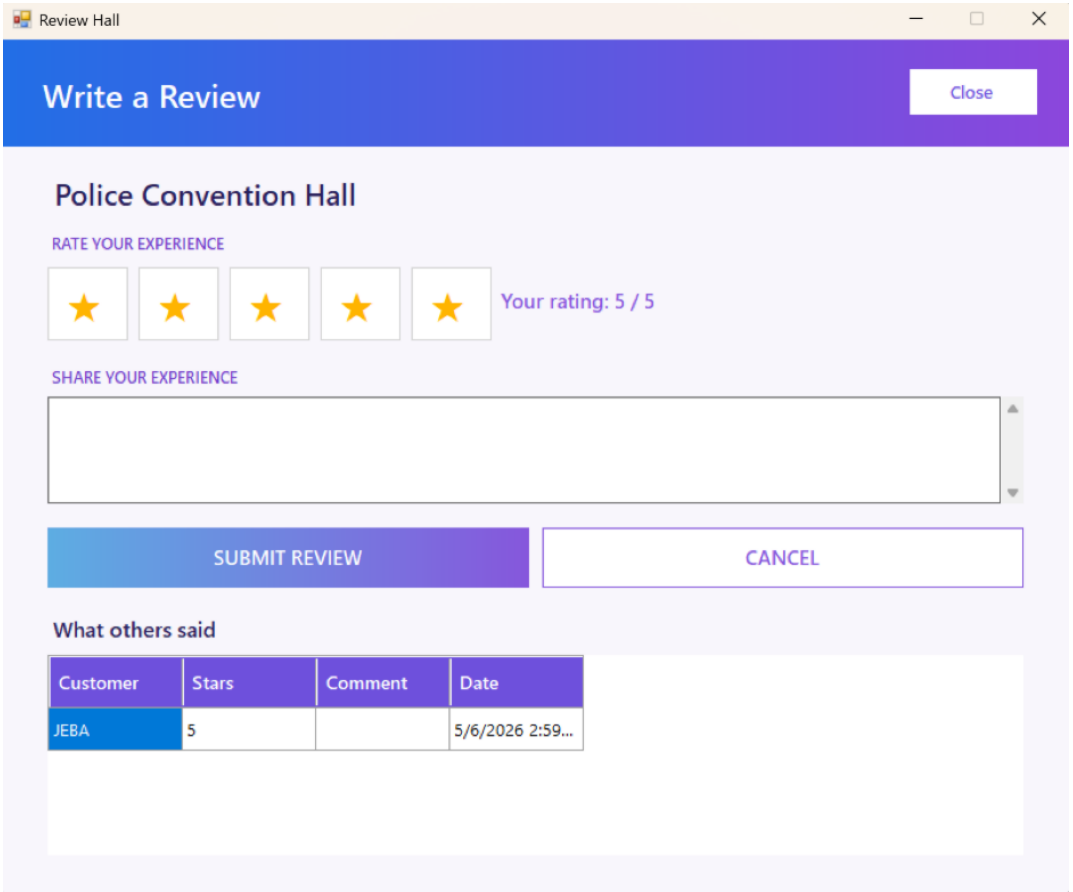
8.6 Booking Page



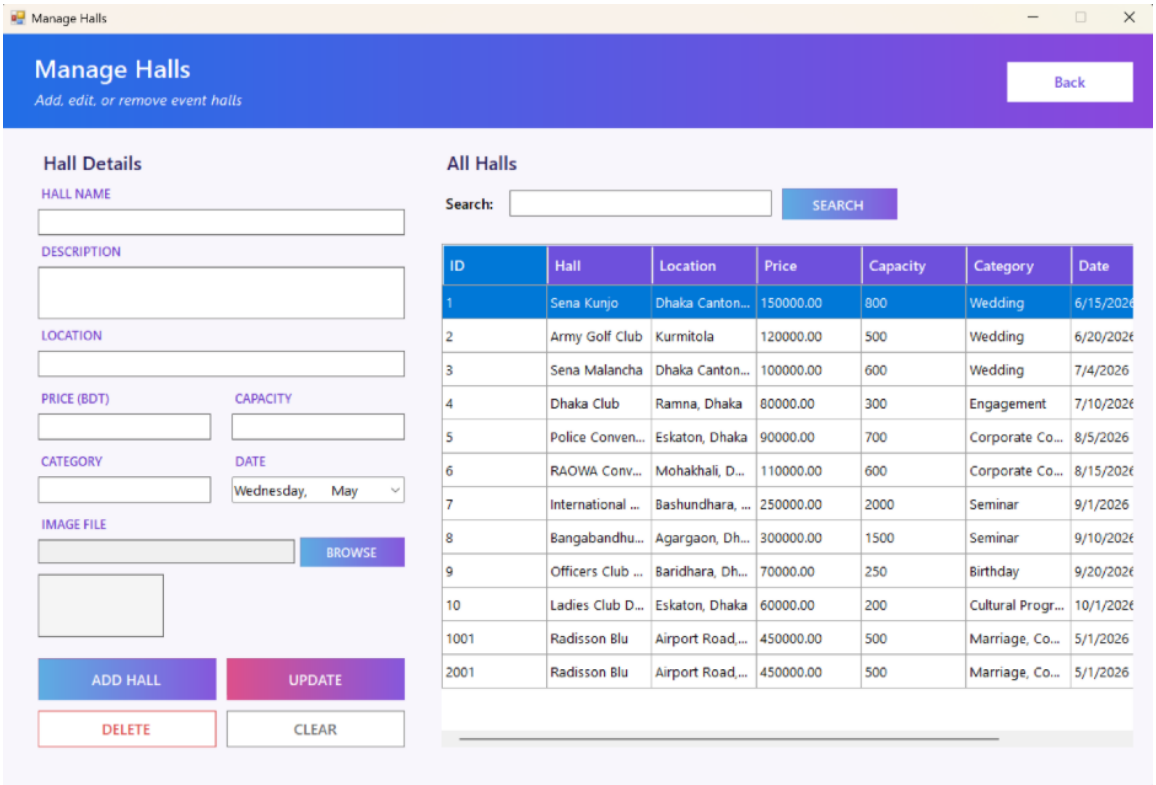
8.7 Payment Page

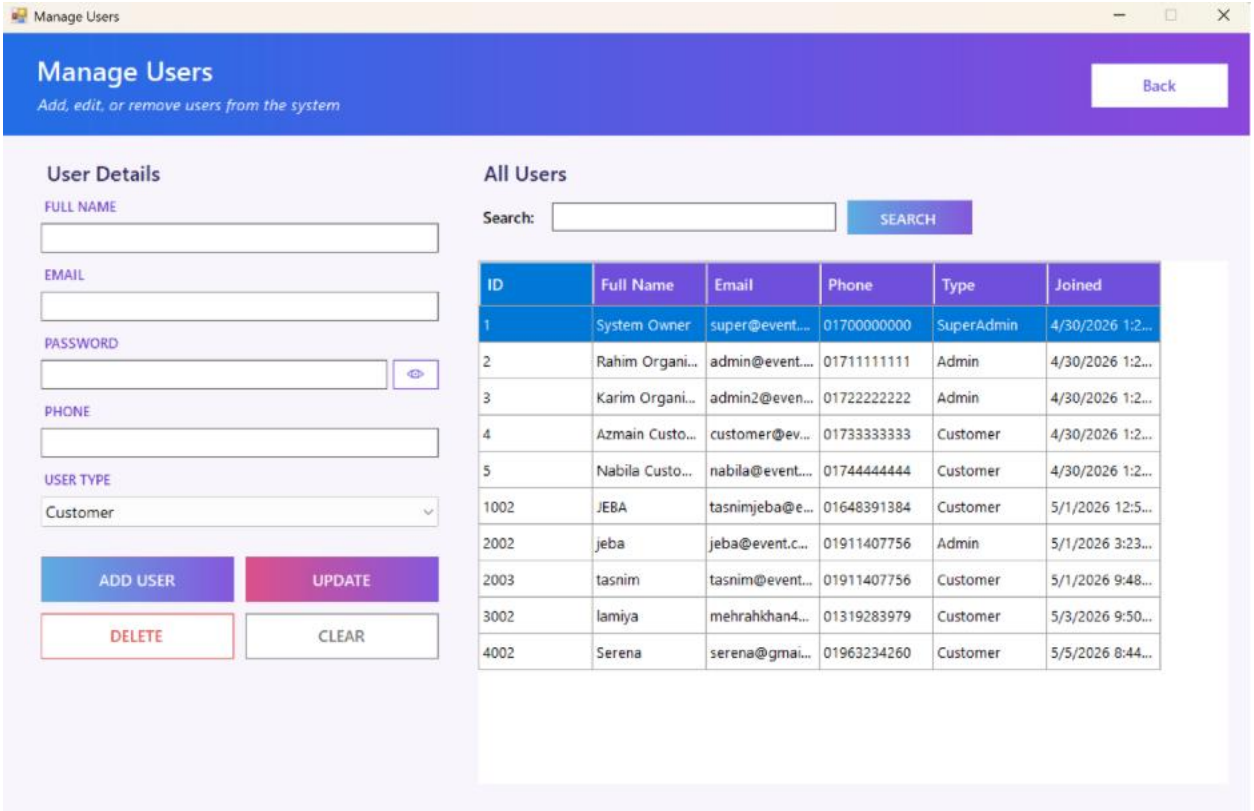


8.8 Review Page

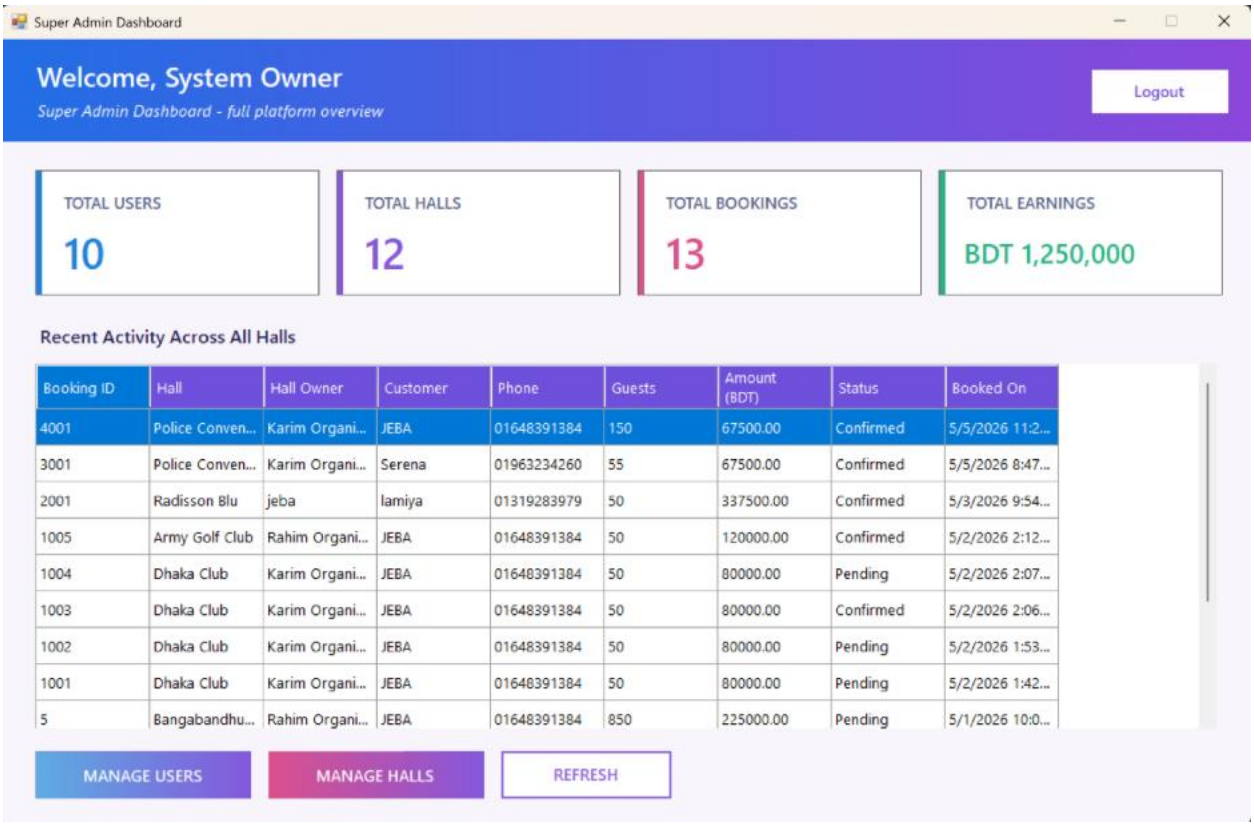

Review Hall
Write a Review
Close
Police Convention Hall
RATE YOUR EXPERIENCE
Your rating: 5 / 5
SHARE YOUR EXPERIENCE
SUBMIT REVIEW
CANCEL
What others said
Customer
Stars
Comment
Date
JEBA
5
5/6/2026 2:59...

8.9 Super Admin Dashboard



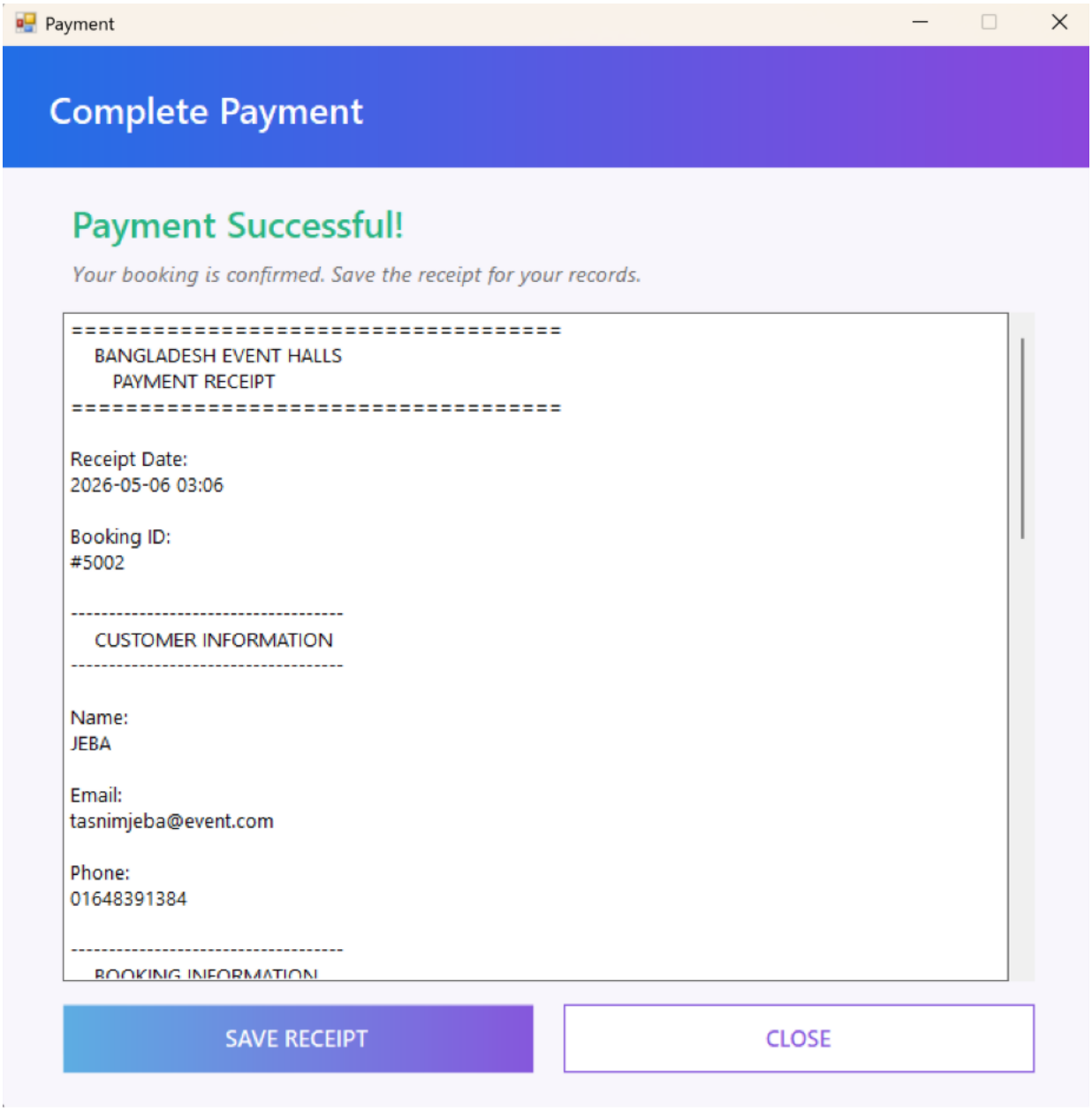


Super Admin Dashboard



Booking ID	Hall	Hall Owner	Customer	Phone	Guests	Amount (BDT)	Status	Booked On
4001	Police Conven...	Karim Organi...	JEBA	01648391384	150	67500.00	Confirmed	5/5/2026 11:2...
3001	Police Conven...	Karim Organi...	Serena	01963234260	55	67500.00	Confirmed	5/5/2026 8:47...
2001	Radisson Blu	jeba	lamiya	01319283979	50	337500.00	Confirmed	5/3/2026 9:54...
1005	Army Golf Club	Rahim Organi...	JEBA	01648391384	50	120000.00	Confirmed	5/2/2026 2:12...
1004	Dhaka Club	Karim Organi...	JEBA	01648391384	50	80000.00	Pending	5/2/2026 2:07...
1003	Dhaka Club	Karim Organi...	JEBA	01648391384	50	80000.00	Confirmed	5/2/2026 2:06...
1002	Dhaka Club	Karim Organi...	JEBA	01648391384	50	80000.00	Pending	5/2/2026 1:53...
1001	Dhaka Club	Karim Organi...	JEBA	01648391384	50	80000.00	Pending	5/2/2026 1:42...
5	Bangabandhu...	Rahim Organi...	JEBA	01648391384	850	225000.00	Pending	5/1/2026 10:0...

8.10 Payment



9. Features and Facilities

The Event Management System provides several features that make hall booking and event management much easier than the traditional way. The most important ones are listed below.

1. 1. Multi-role login (Customer, Admin / Hall Owner, Super Admin) through the Roles table.

2. Realistic Bangladesh halls — Sena Kunja, Army Golf Club, Sena Malancha, Dhaka Club, community and convention centres.
3. Smart hall search by location, event type, price, capacity and date.
4. Booking calendar showing available and busy time slots per hall.
5. Service provider companies stored as records (no login) and assigned to bookings by the admin through EventAssignments.
6. Booking approval workflow controlled by hall owners.
7. Customer reviews with 1–5 star ratings.
8. Multiple payment methods (Cash, Card, bKash, Nagad, Rocket, Bank) with status tracking.
9. Centralised Super Admin control over every table.
10. Microsoft SQL Server database with primary keys, foreign keys and CHECK constraints.

Validation is about checking that the data entered by the user is correct and complete before it is saved into the database. Verification is about checking that the data already inside the system behaves correctly throughout the application. We have applied both ideas in our Event Management System.

Validation Examples

- Empty field check: All required fields (name, email, password, phone, hall name, event date, etc.) must be filled in. The system shows a MessageBox if any field is left empty.
- Email format: A regular expression is used to make sure the email contains an "@" and a valid domain (e.g. user@example.com).
- Phone number: Bangladesh mobile numbers must be 11 digits and start with "01". A simple length and prefix check is applied on the textbox.
- Password match: On the Signup Page, the password and confirm password fields must be exactly equal.
- Numeric fields: Capacity, base price, and number of guests must be positive numbers; the system uses int.TryParse and decimal.TryParse before saving.
- Event date: The event date must not be a date that has already passed and must not be a busy date for the selected hall.
- Rating range: When giving a review, the rating value is forced to be between 1 and 5 using a NumericUpDown control with a CHECK constraint in the database as a backup.

Verification Examples

- Login verification: The system checks the email and password against the matching role table (Customer, HallOwner, EventServiceProvider or SuperAdmin) and only opens the dashboard if both are correct.

- Booking conflict verification: Before saving a new booking, the system checks if the chosen date is already booked for that hall. If yes, the booking is blocked.
- Total price verification: The total price shown on the Payment Page is recalculated on the server side from the database, not just trusted from the client form, so the customer cannot change it.
- Foreign key verification: SQL Server foreign keys make sure no booking can exist without a real customer and a real hall, and no payment can exist without a real booking.
- Status verification: A booking can only move between the values Pending → Approved / Rejected → Completed / Cancelled. Any other status is rejected by the CHECK constraint.
- Try-Catch blocks: Every database call is wrapped in a try-catch so that connection errors or SQL exceptions do not crash the application; they are shown nicely in a MessageBox.

Together, validation and verification make the Event Management System safer, more reliable, and easier to trust for both customers and hall owners.

11. Conclusion

The Event Management System brings the whole hall booking and event service experience of Bangladesh into one simple desktop application. Customers no longer need to call ten different halls one by one or visit them physically just to check a price. They can sit at home, open the application, search halls in places like Mirpur or Banani, see the details of Sena Kunja, Sena Malancha, Army Golf Club or any community centre, and finish a full booking with extra services in just a few minutes.On the other side, hall owners get a clean dashboard to manage their halls and approve only the bookings they really want. Event management groups get a place to show their decoration, catering, photography, stage setup and event planning packages to a much larger audience. The super admin keeps the whole system safe by watching over every customer, owner, provider, booking, payment, and review.Through this project we have learned how to design a real-life database, how to apply the rules of normalization (UNF, 1NF, 2NF, 3NF), how to write proper SQL Server CREATE TABLE queries with primary keys, foreign keys and CHECK constraints, and how to build a clean Windows Forms user interface in C#. We also learned how to apply object-oriented programming concepts in a real desktop application and how to handle validation and verification properly.In the future, this project can be extended into a web-based or mobile application, with online payment gateways like bKash and SSLCommerz fully integrated, automatic email and SMS notifications for booking updates, and an AI-based hall recommendation feature. We believe that with these improvements, the Event Management System can become a real product that is used by event organisers all over Bangladesh.